



EV Charging Station Analysis In India

Project Output Report

Dataset:

EV Charging Stations in India

Total records : 1547

Tools :

- ✓ *Python*
- ✓ *Pandas*
- ✓ *NumPy*
- ✓ *Matplotlib*
- ✓ *Folium*

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Data Understanding

Introduction

This section provides an overview of the EV Charging Station dataset, including its structure, records, columns, and data quality before analysis.

```
op/EV-Charging-Station-Analysis/Python/data_understanding.py
=====
1. FIRST 5 ROWS OF THE DATASET
=====
   name                state  city                address latitude longitude type
0  Neelkanth Star DC Charging Station  Haryana  Gurugram  Neelkanth Star Karnal, NH 44, Gharunda, Kutail...  29.6019  76.9803  12.0
1  Galleria DC Charging Station  Haryana  Gurugram  DLF Phase IV, Sector 28, Gurugram, Haryana 122822  28.4673  77.0818  12.0
2  Highway Xpress (Jaipur-Delhi) DC charging station  Rajasthan  Behror  Jaipur to Delhi Road, Behror Midway, Behror, R...  27.8751  76.2760  12.0
3  Food Carnival DC Charging Station  Uttar Pradesh  Khatauli  Fun and Food Carnival, NH 58, Khatauli Bypass,...  29.3105  77.7218  12.0
4  Food Carnival AC Charging Station  Uttar Pradesh  Khatauli  NH 58, Khatauli Bypass, Bhainsi, Uttar Pradesh...  29.3105  77.7218  12.0
=====
2. LAST 5 ROWS OF THE DATASET
=====
   name                state  city                address latitude longitude type
1542 Tata Power  Kerala  Munnar  Gokulam Park Munnar, Power House Road, South C...  10.0297934  77.045859  7.0
1543 Tata Power  Haryana  Gurgaon  Vatika Town Square II, Sector 82, Sector 82, V...  28.3904593  76.959200  7.0
1544 Tata Power  Haryana  Gurgaon  Zedex TATA, Sec 48, GF-26, NIHO Scottish Mall,...  28.411872  77.040546  7.0
1545 Tata Power  Jammu  Jammu  Le ROI, Jammu, Railway Station, Jammu, Jammu &...  32.7064117  74.879203  7.0
1546 Tata Power  Karnataka  Mangalore  Auto Matrix, Bejai, Manjusha Building, Bejai, ...  12.885716  74.843476  7.0
=====
3. DATASET SHAPE
=====
Rows    : 1547
Columns : 7
```



Output Summary

- ✓ Displayed the first and last five records.
- ✓ Verified that the dataset contains 1,547 rows and 7 columns.
- ✓ Identified available features such as station name, state, city, address, latitude, longitude and charging type.

```
=====
6. STATISTICAL SUMMARY
=====
      longitude      type
count  1.541000e+03  1539.000000
mean   1.105323e+05   9.020793
std    3.064996e+06   4.136436
min    8.058454e+00   6.000000
25%    7.562036e+01   7.000000
50%    7.721257e+01   7.000000
75%    7.847983e+01  11.000000
max    8.510551e+07  24.000000
=====
7. MISSING VALUES
=====
name      0
state     0
city      0
address   40
latitude   6
longitude  6
type      8
dtype: int64
=====
8. DUPLICATE RECORDS
=====
Duplicate Records : 205
```



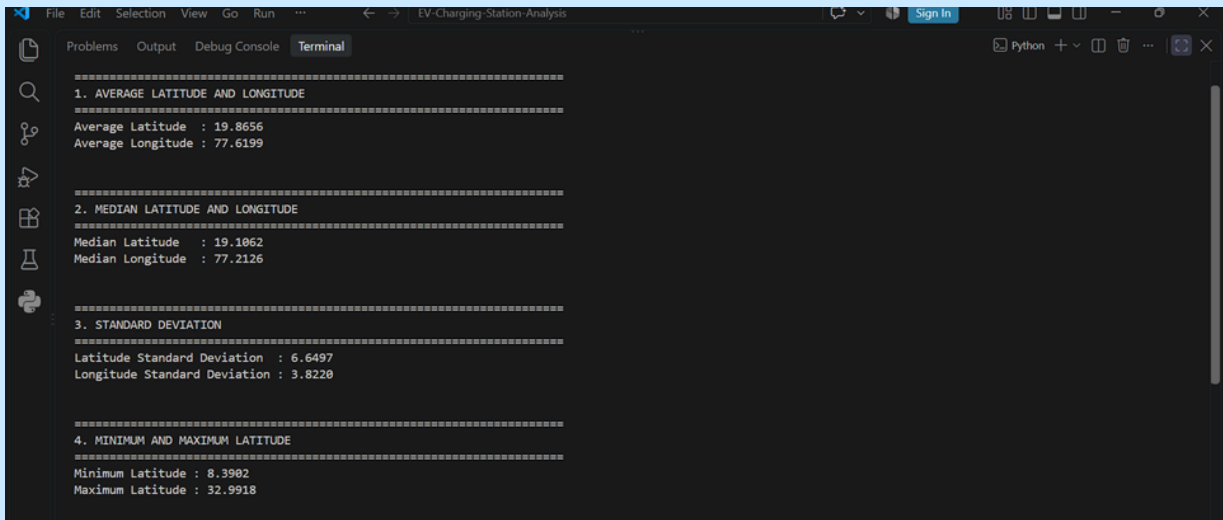
Output Summary

- ✓ Generated statistical summary for numerical columns.
- ✓ Checked missing values to identify incomplete records.
- ✓ Detected duplicate records for further data cleaning before analysis.

NumPy Statistical Analysis

Introduction

This section presents statistical calculations performed using NumPy. It summarizes the geographical distribution of EV charging stations by calculating average, median, standard deviation, and minimum/maximum latitude and longitude values.



```
File Edit Selection View Go Run ... EV-Charging-Station-Analysis Sign In Python + - - - - - x
Problems Output Debug Console Terminal
=====
1. AVERAGE LATITUDE AND LONGITUDE
=====
Average Latitude : 19.8656
Average Longitude : 77.6199

=====
2. MEDIAN LATITUDE AND LONGITUDE
=====
Median Latitude : 19.1062
Median Longitude : 77.2126

=====
3. STANDARD DEVIATION
=====
Latitude Standard Deviation : 6.6497
Longitude Standard Deviation : 3.8220

=====
4. MINIMUM AND MAXIMUM LATITUDE
=====
Minimum Latitude : 8.3902
Maximum Latitude : 32.9918
```

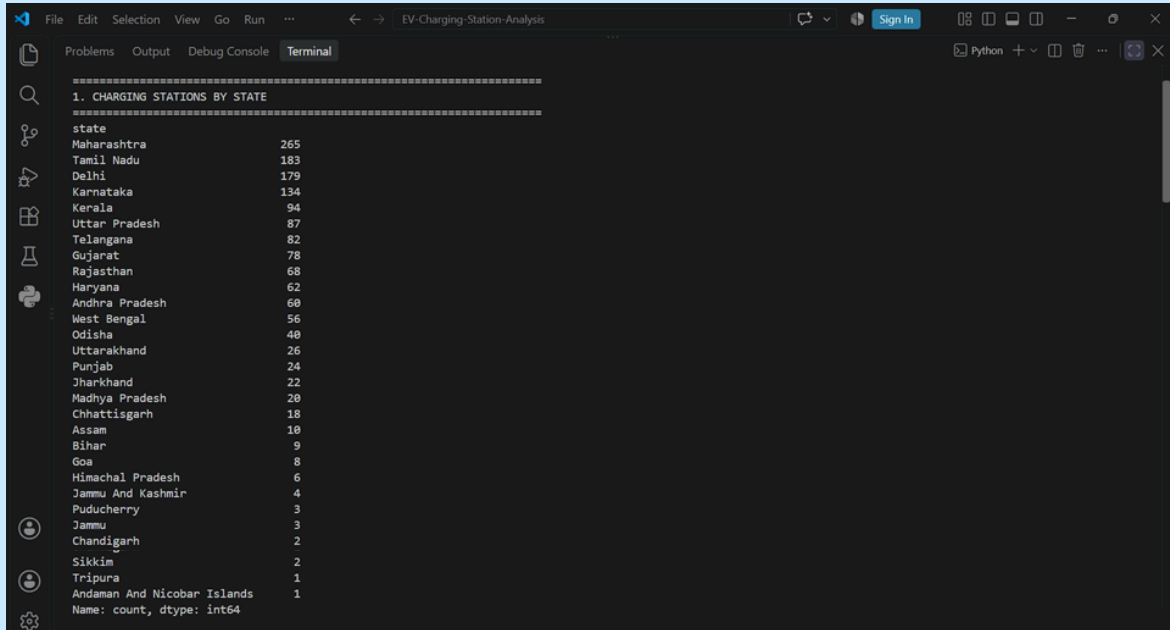
Output Summary

- ✓ Calculated the average latitude and longitude to identify the overall geographical center of EV charging stations in India.
- ✓ Computed the median coordinates to determine the central location while reducing the effect of outliers.
- ✓ Measured the standard deviation to understand how widely the charging stations are geographically distributed.
- ✓ Identified the minimum and maximum latitude values to determine the north-south geographical coverage of the dataset.

Pandas Data Analysis

Introduction

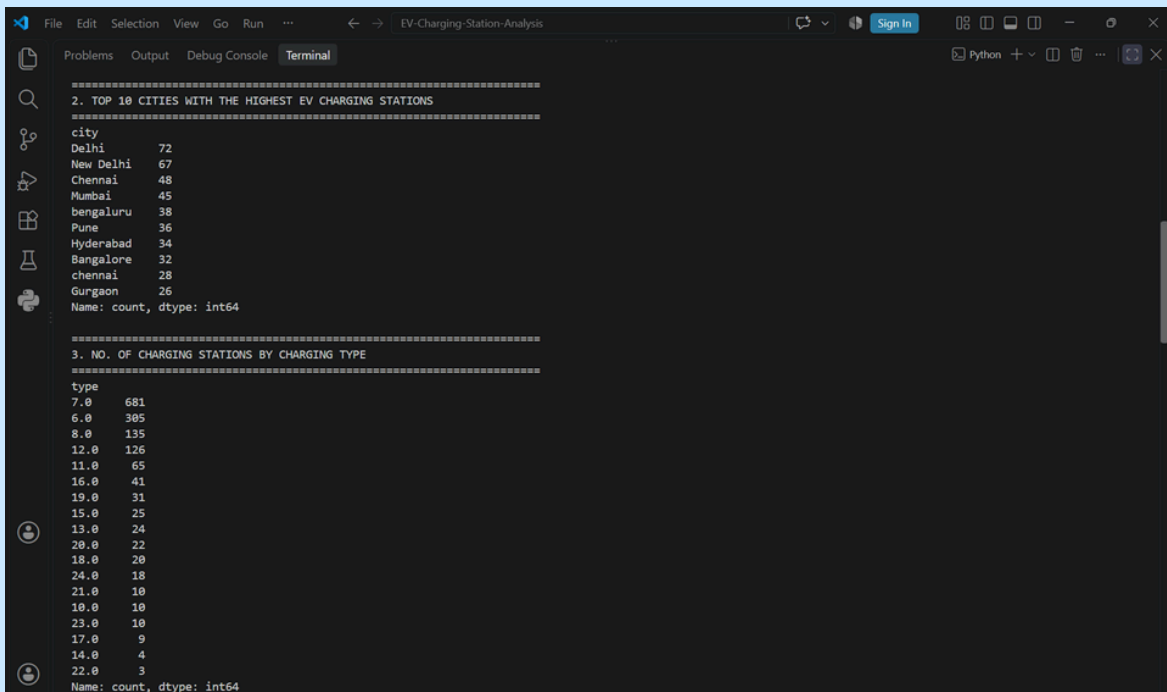
This section analyzes the cleaned EV charging station dataset using Pandas to identify charging station distribution across states, cities, and charging types.



```
=====
1. CHARGING STATIONS BY STATE
=====
state
Maharashtra    265
Tamil Nadu     183
Delhi           179
Karnataka       134
Kerala          94
Uttar Pradesh   87
Telangana       82
Gujarat         78
Rajasthan       68
Haryana         62
Andhra Pradesh  60
West Bengal     56
Odisha          40
Uttarakhand     26
Punjab          24
Jharkhand       22
Madhya Pradesh  20
Chhattisgarh    18
Assam           10
Bihar           9
Goa             8
Himachal Pradesh 6
Jammu And Kashmir 4
Puducherry      3
Jammu           3
Chandigarh      2
Sikkim          2
Tripura         1
Andaman And Nicobar Islands 1
Name: count, dtype: int64
```

Output Summary

- ✓ Analyzed the number of charging stations in each state.
- ✓ Maharashtra recorded the highest number of EV charging stations.
- ✓ Provides a state-wise overview of EV infrastructure.



```
=====
2. TOP 10 CITIES WITH THE HIGHEST EV CHARGING STATIONS
=====
city
Delhi         72
New Delhi     67
Chennai       48
Mumbai        45
bengaluru     38
Pune          36
Hyderabad     34
Bangalore     32
chennai       28
Gurgaon       26
Name: count, dtype: int64

=====
3. NO. OF CHARGING STATIONS BY CHARGING TYPE
=====
type
7.0    681
6.0    305
8.0    135
12.0   126
11.0    65
16.0    41
19.0    31
15.0    25
13.0    24
20.0    22
18.0    20
24.0    18
21.0    10
10.0    10
23.0    10
17.0     9
14.0     4
22.0     3
Name: count, dtype: int64
```



Output Summary

- ✓ Delhi has the highest number of EV charging stations among cities.
- ✓ Charging type 7.0 is the most common in the dataset.
- ✓ Top cities show a higher concentration of charging stations in major urban areas.

```
File Edit Selection View Go Run ... EV-Charging-Station-Analysis Sign In
Problems Output Debug Console Terminal
=====
4. TOP 10 STATES WITH THE HIGHEST EV CHARGING STATIONS
=====
state
Maharashtra    265
Tamil Nadu     183
Delhi          179
Karnataka      134
Kerala         94
Uttar Pradesh  87
Telangana      82
Gujarat        78
Rajasthan      68
Haryana        62
Name: count, dtype: int64
=====
5. NUMBER OF UNIQUE STATES
=====
Unique States : 29

=====
6. NUMBER OF UNIQUE CITIES
=====
Unique Cities : 362

=====
7. NUMBER OF UNIQUE CHARGING TYPES
=====
Unique Charging Types : 18
```



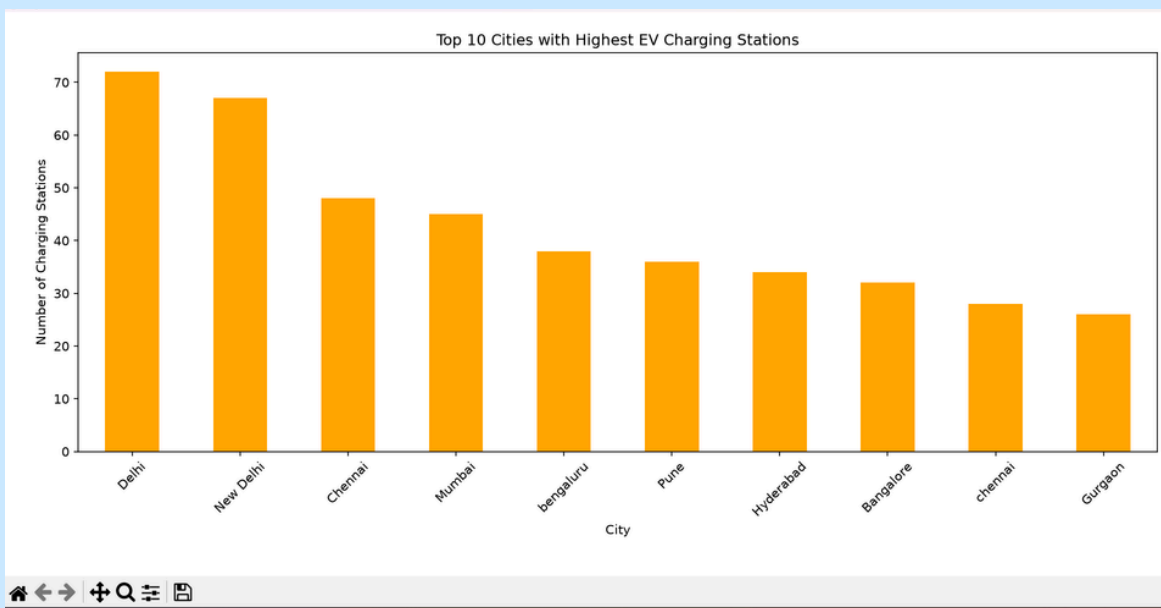
Output Summary

- ✓ Maharashtra has the highest number of charging stations among states.
- ✓ The dataset covers 29 unique states, 362 unique cities, and 18 charging types.
- ✓ EV charging infrastructure is concentrated more in major states.

Data Visualization & Insights

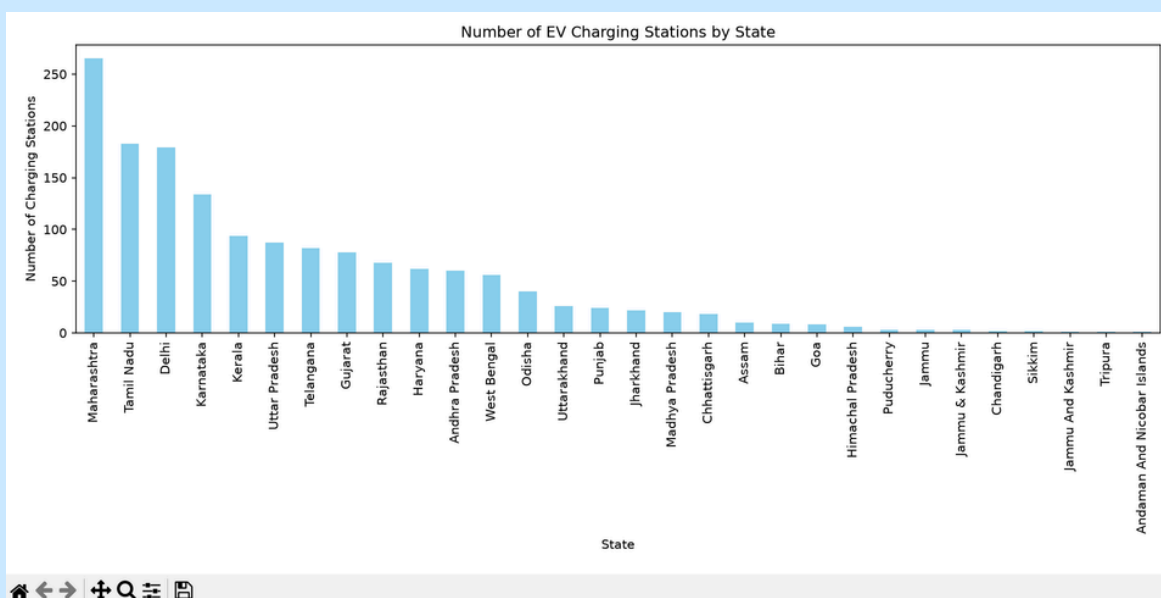
Introduction

This section presents visual insights from the EV Charging Stations dataset using different charts. The visualizations help understand the distribution of charging stations across major cities and states, along with the most frequently recorded charging station types. These charts make the key patterns and trends in the dataset easier to understand.



Output Summary

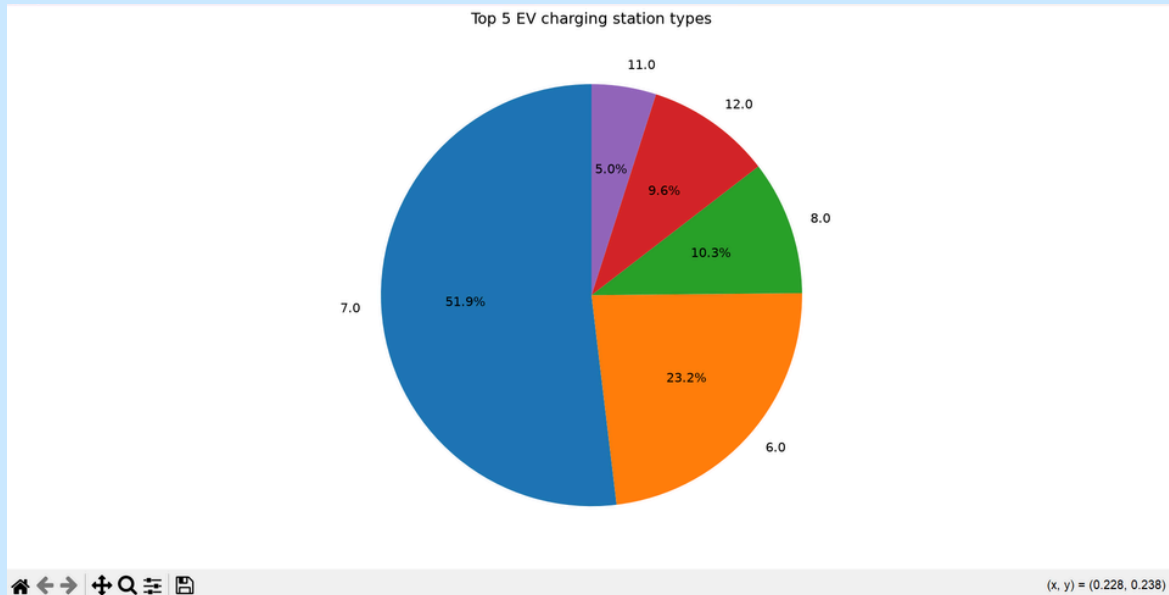
- ✓ The chart shows the top 10 cities with the highest number of EV charging stations.
- ✓ Delhi has the highest number of charging stations among the listed cities.
- ✓ Major cities show a higher concentration of EV charging infrastructure.





Output Summary

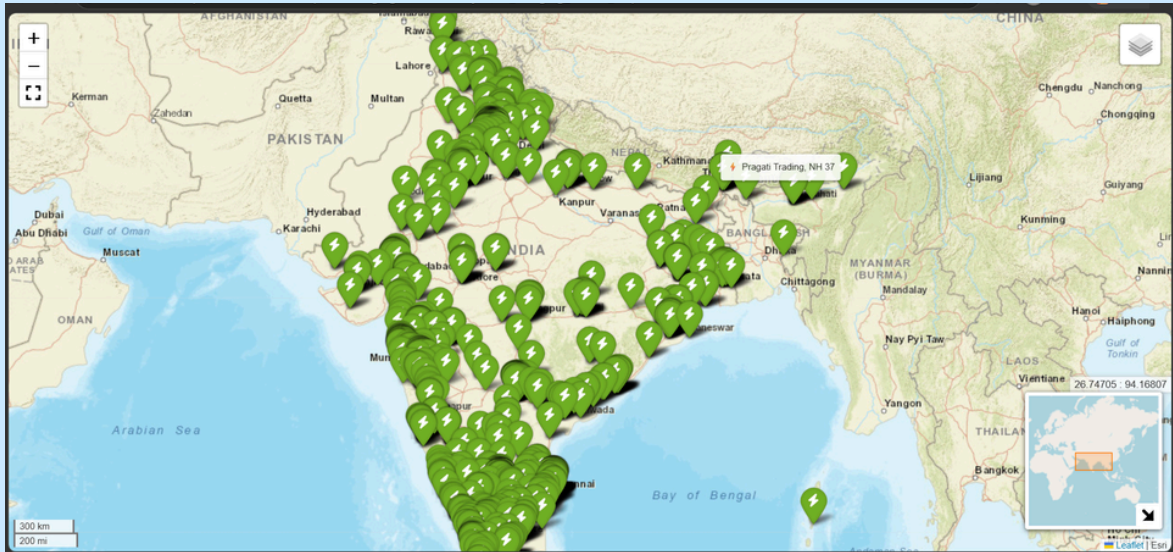
- ✓ The chart shows the state-wise distribution of EV charging stations in India.
- ✓ Maharashtra has the highest number of charging stations.
- ✓ The distribution shows that EV charging infrastructure is more concentrated in a few major states.



Output Summary

- ✓ The pie chart shows the distribution of the five most frequently recorded charging types.
- ✓ Charging Type 7.0 has the largest share among the top five types.
- ✓ The chart provides an overview of the most commonly recorded charging types in the dataset.

Interactive EV Charging Station Map



Output Summary

- ✓ The map visualizes the geographical distribution of EV charging stations across India.
- ✓ Charging stations are widely distributed across different regions, with higher concentration in major cities and states.
- ✓ Interactive markers provide station details such as name, city, state, address, and charging type.
- ✓ The map helps understand the overall geographical spread of EV charging infrastructure in India.